

LESSON 114 (0.5 GROUND and 1.3 DUAL)

Lesson Objective

The purpose of this lesson is to introduce normal and crosswind takeoffs as their understanding of how to control the aircraft should allow for this by this point. In addition, ground reference maneuvers will be introduced to the applicant to increase their ability in correcting for wind based on a ground reference. The applicant will continue familiarizing themselves with radio communication procedures with instructor guidance

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the readings as needed.
 - Chapter 07 of the Airplane Flying Handbook- Ground Reference Maneuvers.
 - Ask the students if they have any questions about the procedures from the maneuver's manual.
 - Normal and Crosswind takeoffs.
 - Ground Reference maneuvers.
 - Have the student draw out turns around a point and s-turns to the best of their understanding.
- Student questions.

References

- Advisory Circular. ([Search Tool](#))
 - Stall and Spin Awareness Training. ([AC 61-67](#))
- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 05 - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
 - Chapter 07 - Ground Reference Maneuvers.
- Warrior Maneuvers Manual.

Instructor guidance

- This lesson introduces ground reference maneuvers for the first time in the aircraft. Before departing, have the student draw out turns around a point and S-turns as instructed in the briefing. A student who cannot sketch the ground track and explain the bank variation on the ground will struggle significantly in the air — address any gaps before the flight.
- For every ground reference maneuver in this lesson, strictly follow the telling and doing technique: demonstrate first with the student narrating ("instructor does, student says"), then have the student perform it with narration ("student says, student does"). Do not skip the demonstration phase, even if the student feels confident after the ground brief.
- Wind assessment is the foundation of every ground reference maneuver. Before entering any maneuver, require the student to identify the wind direction using available cues — smoke, water movement, the PFD groundspeed, or the windsock. Make this a non-negotiable step in the entry procedure every time.

- During turns around a point, watch closely for students who fix their bank angle rather than varying it with groundspeed. The most common error is entering with a bank that is too shallow on the downwind side. Allow the student to see the radius widen before prompting a correction — the visual feedback is more effective than telling them preemptively.
- For S-turns, the most challenging moment is the crossover — rolling out of one turn and immediately rolling into the opposite turn on the reference line. Brief this transition explicitly on the ground and demonstrate it clearly in the air before the student attempts it.
- The rectangular course is a direct precursor to the traffic pattern. Make this connection explicit during the briefing and reinforce it in the air. Students who understand why they are flying a rectangular course — because it mirrors the pattern — will engage with the maneuver more meaningfully.
- Division of attention is the overarching skill being developed in ground reference maneuvers — the student must simultaneously manage aircraft control, ground track, wind correction, and traffic scanning. If the student becomes fixated on the reference point and neglects altitude or coordination, prompt them immediately and use it as a teaching moment about situational awareness.
- Keep an eye on altitude throughout all ground reference maneuvers. Students new to these maneuvers often allow altitude to creep up or down during bank transitions. Reinforce the completion standard of ± 150 ft early so the student has a clear performance target.

Lesson Content (Ground)

Normal and Crosswind Takeoff

- Taxi into takeoff position when cleared for takeoff by ATC and verify magnetic heading matches assigned takeoff runway heading.
- Smoothly and continuously apply takeoff power.
- Check static RPM (approximately 2330 RPM) and verify airspeed rising.
- Use rudder as necessary to maintain directional control and ailerons (into the wind) to aid with crosswind correction.
- When the flight controls become effective, gradually apply back stabilator pressure to attain a lift-off pitch attitude.
- Ensure that ailerons are neutral prior to rotating.
- Note the position of the airplane's nose in relation to the horizon and adjust as necessary to attain a VY climb pitch attitude.
- At 1,000 feet AGL, lower the nose, approximately an inch, to attain a cruise climb pitch attitude and complete the climb check.

Turns Around a Point

- The student should follow along with the Airplane Flying Handbook chapter 7.
- Ground reference maneuver that consists of a 360 degrees (x2) constant radius turn around a ground-based reference point.
- Select a prominent ground-based reference point.
 - Mention obstacle avoidance and landing point in case of emergency.
- Enter on the downwind at 1,000 feet.

- Use smoke, water, PFD, groundspeed, etc. to determine where the winds are coming from.
- Mention how close/far away from the point the entry should be.
- Emphasize space and time to set up!
- Higher groundspeed requires a steeper bank and vice-versa. The bank angle will be adjusted throughout the maneuver.
- Objectives of the maneuver:
 - Adjusting the bank angle accordingly in correlation with the groundspeed.
 - Maintaining a specific distance between the airplane and the reference point.
 - Division of attention.
 - Compensating for wind drift.
- Common errors:
 - Failure to clear the area.
 - Failure to enter the maneuver at a constant altitude and airspeed.
 - Failure to maintain altitude during the maneuver.
 - Failure to assess wind direction.
 - Failure to maintain a constant radius turn.
 - Failure to maintain airplane coordination.

S-Turns

- The student should follow along with the Airplane Flying Handbook chapter 7.
- Ground reference maneuver in which the airplane's ground track forms two opposite and equal half-circles on each side of a ground-based straight-line reference.
- Select a prominent straight-line ground-based reference point (perpendicular to the wind).
 - Mention obstacle avoidance and landing point in case of emergency.
- Enter on the downwind at 1,000 feet.
- Use smoke, water, PFD, groundspeed, etc. to determine where the winds are coming from.
- Emphasize space and time to set up!
- throughout the maneuver.
- Objectives of the maneuver:
 - Adjusting the bank angle accordingly in correlation with the groundspeed.
 - Maintaining a specific distance between the airplane and the reference point.
 - Division of attention.
 - Compensating for wind drift.
- Common errors:
 - Failure to clear the area.
 - Failure to enter the maneuver at a constant altitude and airspeed.
 - Failure to maintain altitude during the maneuver.
 - Failure to assess wind direction.
 - Failure to maintain a constant radius turn.
 - Failure to maintain airplane coordination.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:

- Personal minimums.
- FIT minimums.
- Aircraft.
- Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Dual)

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all the checklists.
- Student should be able to perform the “passenger briefing” and the “engine fire during start brief”.

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Operation of the Communications and Navigation Units

- Ensure that the student understands how to properly use COMS and NAV units.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command’s responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Radio Communication Procedures and ATC Phraseology

- Students should understand and demonstrate:
 - Proper radio setup and frequency management
 - Initial call-up format (who you are calling, who you are, where you are, what you want)
 - Accurate readback of clearances and instructions (runway assignments, hold short instructions, altitudes, headings, squawk codes)
 - Use of standard terminology (e.g., “affirmative,” “negative,” “standby,” “unable”)
 - Avoid nonstandard or casual language
 - Maintaining situational awareness while communicating
- The student should be able to conduct radio calls in the practice area.
- The instructor should assist as needed.
- Review emphasis areas from lesson 112.

Normal and Crosswind Takeoff

- The student should conduct the takeoff.
- Ensure the runway entry check and the lineup check are completed.
- Ensure proper crosswind correction is maintained.
- Have the student vocalize the runway line-up checklist and verify all line items.
- Ensure the student establishes full power with one hand on the throttle and one hand on the yoke.
- Verify the student is inputting proper crosswind correction.
- Ensure the student points out the windsock and corrects accordingly.
- Once off the ground, TRIM for desired airspeed!

Review Previous Exercises as Required

- Use the appropriate lesson plan depending on what needs to be reviewed.

Turns Around a Point

- First completion of the maneuver will be a demonstration.
 - Instructor does, student says.
 - Verify correct procedure.
 - Points of Emphasis.
 - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
 - Bank variation.
 - Choosing an adequate and safe visual reference point.
- Second completion of the maneuver will be performed by the student
 - Student says, student does.
 - Verify correct procedure.
 - Points of Emphasis.
 - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
 - Bank variation.
 - Choosing an adequate and safe visual reference point.
 - Provide feedback as needed throughout the maneuver.

S-turns

- First completion of the maneuver will be a demonstration.
 - Instructor does, student says.
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 - Choosing an adequate and safe visual reference point.
 - Provide feedback as needed throughout the maneuver.

Rectangular Course

- Find 4 points creating a large rectangle.
- Enter the downwind leg at a 45-degree angle and 2500 ft.
- Practice non-specific (base) vs specific (final) aiming point techniques.
- Initiate a go-around on the final leg at 1500 ft.
- First completion of the maneuver will be a demonstration.
 - Instructor does, student says.
 - Verify correct procedure.
 - Points of Emphasis.
 - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
 - Bank variation.
 - Choosing an adequate and safe visual reference point.
- Second completion of the maneuver will be performed by the student
 - Student says, student does.
 - Verify correct procedure.
 - Points of Emphasis.
 - Entry procedures: downwind, sufficient space to set up, and proper spacing from point.
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 - Choosing an adequate and safe visual reference point.
 - Provide feedback as needed throughout the maneuver.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 115.
- Assign homework:
 - Review the Warrior Maneuvers Manual:

- Power-off stall.
- Power-on stall.
- Turns around a point.
- Review the Airplane Flying Handbook:
 - Chapter 07 (ground reference maneuvers).

Completion Standards

This lesson is complete when the applicant can:

1. Demonstrate proper sequential checklist procedures for all normal procedures without instructor guidance.
2. Correctly perform previously learned maneuvers with minimal instructor guidance.
3. Conduct normal and crosswind takeoffs with instructor guidance.
4. Describe the wind correction required in every ground reference maneuver.
5. During ground reference maneuvers:
 - a. Use correct wind correction with instructor guidance.
 - b. Complete each ground reference maneuver twice using different reference points each time.